THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Classical Optimization Algorithms for Lasso Regression

Four solvers · One objective · A common accuracy target

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Opening: ZHANG Kejia | 30-minute presentation + 10-minute Q&A

github.com/Caiyangxiao/COMP6704-Group9-Project

Our Work and Presentation Plan

Four implementations answer one shared optimization question.

I Background

CAI Yangxiao

Formulation, convexity and sparse optimality

II Algorithms

XUE Pengyu

ISTA, FISTA, cyclic CD, ADMM and a common protocol

III Results

ZHOU Xiyu

Accuracy, runtime, sensitivity and conditional conclusions

Introduction and work overview: ZHANG Kejia | Experimental design and evaluation: all four members

Group 9 four-solver report, Sections 1–6; member contributions confirmed by the group.

PART I

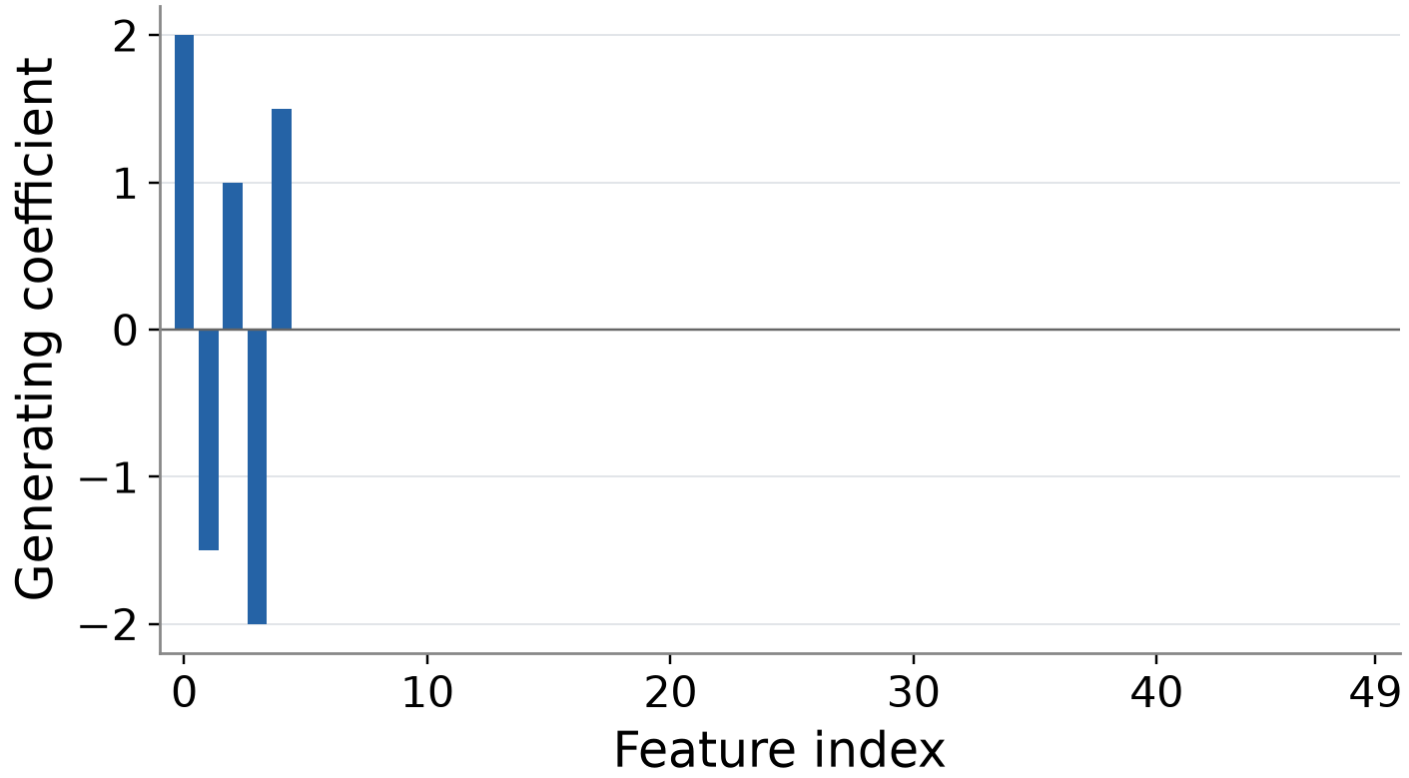
Background and formulation

CAI Yangxiao

Student ID: 26023041R

Why Study Lasso?

Sparse regression makes optimization and feature selection visible.



50 candidate features

5 generating variables

Convex + nonsmooth

A tractable model with algorithmic choices

Known generating coefficients give a direct check of support recovery and shrinkage.

Report Sections 1 and 4.1; saved synthetic-generation specification. Bars show the generating coefficients.

The Same Lasso Objective for Every Solver

Every solver minimizes the same normalized Lasso objective.

$$\min_{w \in \mathbb{R}^d} F(w) = \frac{1}{2n} \|Xw - y\|_2^2 + \lambda \|w\|_1$$

Squared training loss

Fits centered responses

X: n × d, w: d coefficients

L1 penalty $\lambda\|w\|_1$

Encourages sparse coefficients

λ controls regularization

Center and scale using training statistics; restore an unpenalized intercept for prediction.

Report Section 2.1, Eq. (1); Tibshirani (1996).

Related Regression Penalties

Changing the penalty changes the estimator.

Model	Penalty	Main distinction
Ridge	$\alpha\|\|w\|\|_2^2 / 2$	Usually dense coefficients
Lasso	$\lambda\|\|w\|\|_1$	Exact zeros through thresholding
Elastic net	$\lambda\|\|w\|\|_1 + \alpha\|\|w\|\|_2^2 / 2$	Sparsity and correlated-variable grouping
Adaptive Lasso	$\lambda \sum_j \omega_j \|w_j\|$	Fixed weighted thresholds

The benchmark fixes unweighted Lasso and compares how four algorithms solve it.

Report Section 1.2 and Table 1; Hoerl & Kennard (1970), Tibshirani (1996), Zou & Hastie (2005), Zou (2006).

Convexity and Sparse Optimality

Convexity supports global optimization; subgradients permit zeros.

$$0 \in \nabla g(w^\star) + \lambda\, \partial \|w^\star\|_1$$

Convex objective

$\nabla^2 g(w) = X^\top X / n \succeq 0$

$\lambda\|w\|_1$ is convex

Every local minimum is global.

Coordinate optimality

$w^*_j \neq 0$: $\nabla_j g = -\lambda\, \mathrm{sign}(w^*_j)$

$w^*_j = 0$: $|\nabla_j g| \leq \lambda$

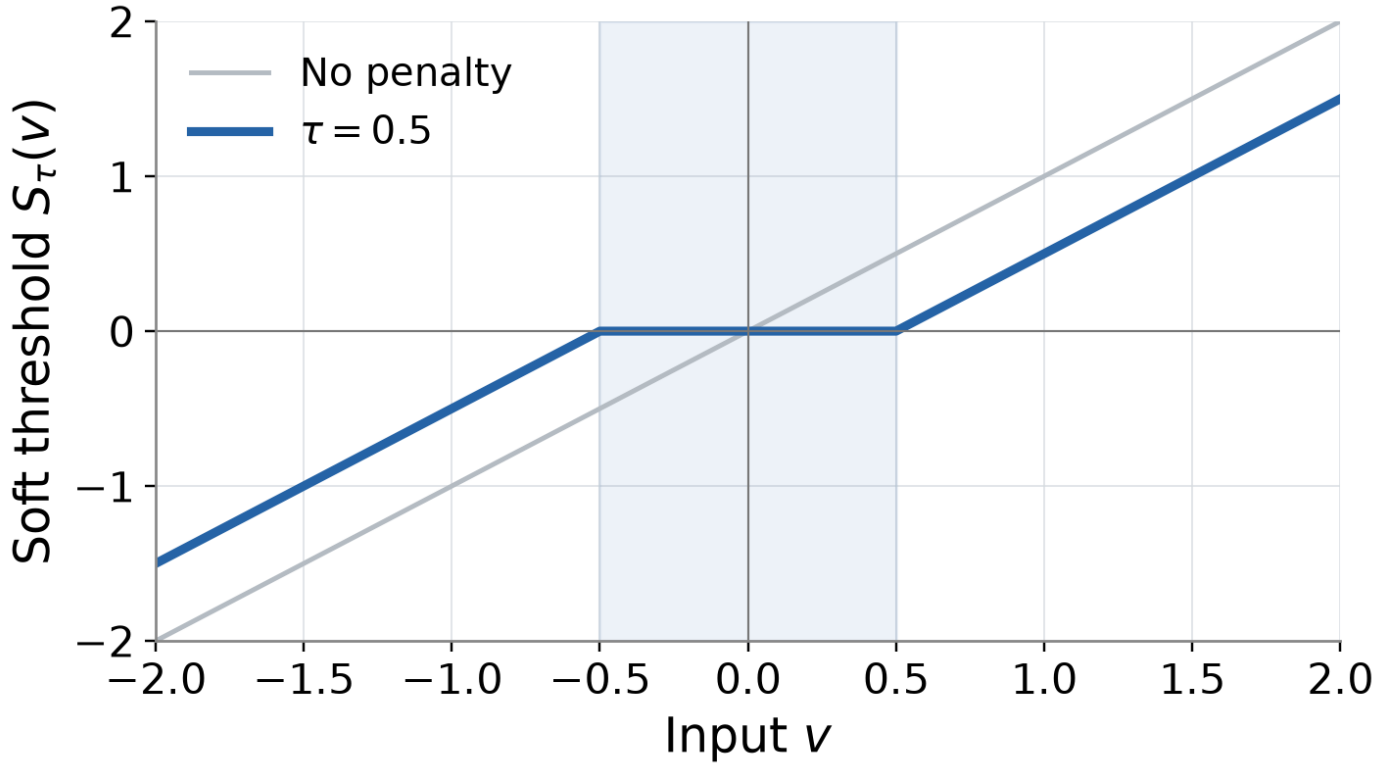
$\partial|0| = [-1, 1]$

Coefficient uniqueness requires additional structure; optimality and true-support recovery are separate checks.

Report Section 2.2; Boyd & Vandenberghe (2004), Tibshirani (2013).

Soft Thresholding

Soft thresholding sets small inputs to zero and shrinks larger ones.



$$S_\tau(v) = \text{sign}(v)\max\{|v| - \tau, 0\}$$

Input	τ = 0.5
1.2	0.7
−0.8	−0.3
0.2	0

The same proximal operation appears in ISTA, FISTA, coordinate descent and ADMM.

Report Section 2.3, Eq. (5); Parikh & Boyd (2014).

Research Questions and Solver Map

Fix the model, then compare update mechanisms and computation.

ISTA

Gradient step
+ threshold

Implemented by
ZHOU Xiyu

FISTA

Extrapolation
+ threshold

Implemented by
XUE Pengyu

CD

Exact scalar
minimization

Implemented by
ZHANG Kejia

ADMM

Linear solve
+ split updates

Implemented by
CAI Yangxiao

Same data, preprocessing, λ and final stationarity target across all four implementations.

Report Sections 3.1–3.4 and confirmed author contributions; classical methods, no new algorithm claimed.

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PART II

Algorithms and experimental design

XUE Pengyu

Student ID: 26022951R

ISTA: Gradient Step, Then Threshold

ISTA takes a fitting step, then applies the L1 proximal map.

$$w^{k+1} = S_{\lambda/L}\left(w^k - \frac{\nabla g(w^k)}{L}\right)$$

1 Current iterate

w^k

2 Gradient step

$\eta = 1/L$

3 Threshold

$\tau = \eta\lambda$

O(nd) dense work per monitored round; a nonincreasing objective with a valid step size.

Report Section 3.1; Daubechies et al. (2004), Beck & Teboulle (2009). Implementation: ZHOU Xiyu.

FISTA: Add Extrapolation

FISTA adds momentum through an extrapolated evaluation point.

$$w^k = S_{\lambda/L}\left(z^k - \frac{\nabla g(z^k)}{L}\right)$$

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}$$

$$z^{k+1} = w^k + \frac{t_k - 1}{t_{k+1}}(w^k - w^{k-1})$$

Acceleration

$O(k^{-2})$ objective-error bound

Standard FISTA can oscillate.

No restart is used here.

A stronger objective-error guarantee does not determine elapsed time to the common residual threshold.

Report Section 3.1; Beck & Teboulle (2009). Implementation: XUE Pengyu.

Cyclic Coordinate Descent

Cyclic CD solves one coefficient exactly using the latest residual.

$$r^{(j)} = y - \sum_{\ell \neq j} x_\ell w_\ell, \qquad w_j \leftarrow \frac{S_\lambda(x_j^\top r^{(j)}/n)}{\|x_j\|_2^2/n}$$

Residual reuse

$r^{(j)} = r + x_j w_j$

Update w_j, then $r = r^{(j)} - x_j w_j$

One complete sweep

Visit j = 1, …, d

O(n) per coordinate; O(nd) per sweep

A small sweep count can still contain many sequential scalar updates.

Report Section 3.2; Friedman et al. (2010), Wu & Lange (2008). Implementation: ZHANG Kejia.

ADMM: Split the Loss and Penalty

ADMM separates squared loss and the L1 penalty through x = z.

$$(X^\top X/n + \beta I)x^{k+1} = X^\top y/n + \beta(z^k - u^k)$$

$$z^{k+1} = S_{\lambda/\beta}(x^{k+1} + u^k)$$

$$u^{k+1} = u^k + x^{k+1} - z^{k+1}$$

β > 0 is the algorithmic penalty. λ controls the Lasso model.

Initialize x, z, u at zero. Return w = z. Main benchmark: β = 1.

The matrix, threshold and objective all use the same 1/(2n) loss normalization.

Report Section 3.3, Eqs. (14)–(16); Boyd et al. (2011). Implementation: CAI Yangxiao.

ADMM Implementation and Correct Evaluation

ADMM reuses a factorization within each cold solve.

Prepare once

Form $X^TX/n + \beta I$

Cache a Cholesky factor

Include this cost in total time.

Dense preparation: $O(nd^2 + d^3)$

Repeat the split updates

Triangular solve, threshold, dual step

Monitor the same F(z) and R(z)

Per monitored round: $O(nd + d^2)$

Extra matrix storage: $O(d^2)$

No factorization is shared across timing repetitions, instances or penalty settings.

Report Sections 3.3–3.4; current dense Cholesky implementation and common monitoring protocol.

Algorithm Trade-offs and Fair Comparison

Rounds expose update counts; elapsed time includes their different costs.

Method	Core operation	Objective	Monitored work
ISTA	Vector thresholding	Monotone	O(nd)
FISTA	Extrapolation + threshold	May oscillate	O(nd)
Cyclic CD	d exact scalar updates	Monotone	O(nd) / sweep
ADMM	Cached solve + split updates	May oscillate	$O(nd + d^2)$

All comparisons use the same returned-vector stationarity measure and cold total time.

Report Section 3.4 and Table 2; no universal cyclic-CD or ADMM rate is assigned.

Data and Controlled Studies

Saved inputs and paired designs isolate the effects we vary.

Study	Samples / features	Instances and control
Synthetic	300 / 50; train 240	Seeds 0–2; 5 active variables
Diabetes	442 / 10; train 353	One saved split; true support unknown
Correlation	600 / 60; train 480	$\rho = 0, 0.5, 0.9, 0.99$; paired seeds 10–12
Dimension	1,000 / 50–600; train 800	$d = 50, 100, 300, 600$; paired prefixes

Train-only preprocessing and fixed splits are shared by every solver on each problem.

Report Section 4.1; CSV hashes, controlled NPZ inputs and split indices are retained.

One Common Stopping Rule and Timing Protocol

Stop at the same stationarity accuracy on the returned coefficients.

$$G_L(w) = L\Big[w - S_{\lambda/L}\Big(w - \frac{\nabla g(w)}{L}\Big)\Big]$$

$$R(w) = \frac{\|G_L(w)\|_\infty}{\max\{1, \|X^\top y/n\|_\infty\}} \le 10^{-6}$$

- $R(w) \le 10^{-6}$; maximum 10,000 complete rounds.
- Five interleaved timings; report each instance's median.
- Total includes preparation and monitoring; shared spectral setup is recorded separately.

R = 0 is KKT stationarity. A small R is not an objective-error upper bound.

Report Section 4.2, Eqs. (18)–(19). ADMM uses w = z; primal/dual split residuals are extra diagnostics.

Validation and Reproducibility

Validate the normalization, then preserve the evidence needed to rerun it.

6 correctness tests passed

Orthogonal closed form

Normalized ADMM first update

Independent reference across β

Zero threshold and rank-deficient design

560 primary + 45 sensitivity runs

All reached $R \le 10^{-6}$

Raw timings, coefficients and traces

Fixed inputs, configs and source hashes

11 figure groups from saved data

All four methods were rerun on one host; historical timings are preserved separately.

Report Section 4.3; tests/test_optimizers.py and results/2026-11-06-four-solvers/.

PART III

Results and conclusions

ZHOU Xiyu

Student ID: 26025265R

Nearly Identical Final Lasso Solutions

Converged solvers give nearly identical fitted results.

Solver	Synthetic test MSE	Diabetes test MSE	Nonzero coefficients
ISTA	0.396 ± 0.139	2966.935	5 on every main input
FISTA	0.396 ± 0.139	2966.936	5 on every main input
CD	0.396 ± 0.139	2966.937	5 on every main input
ADMM	0.396 ± 0.139	2966.937	5 on every main input

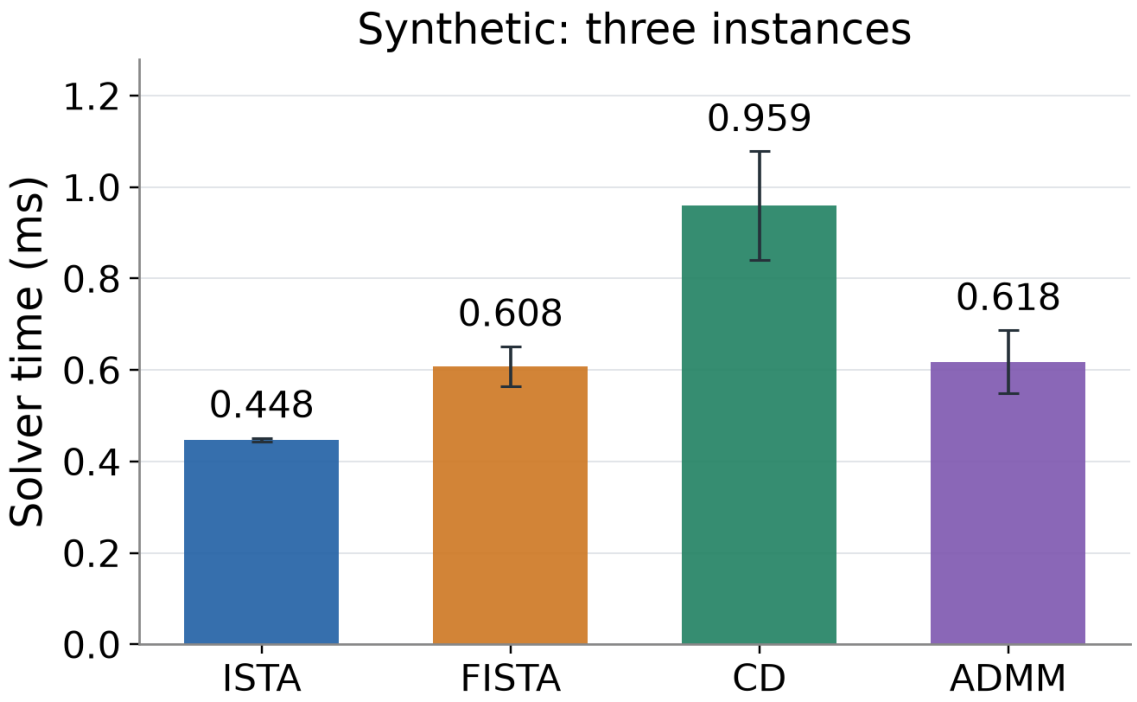
Maximum main within-problem relative objective spread: 3.17×10^{-12}

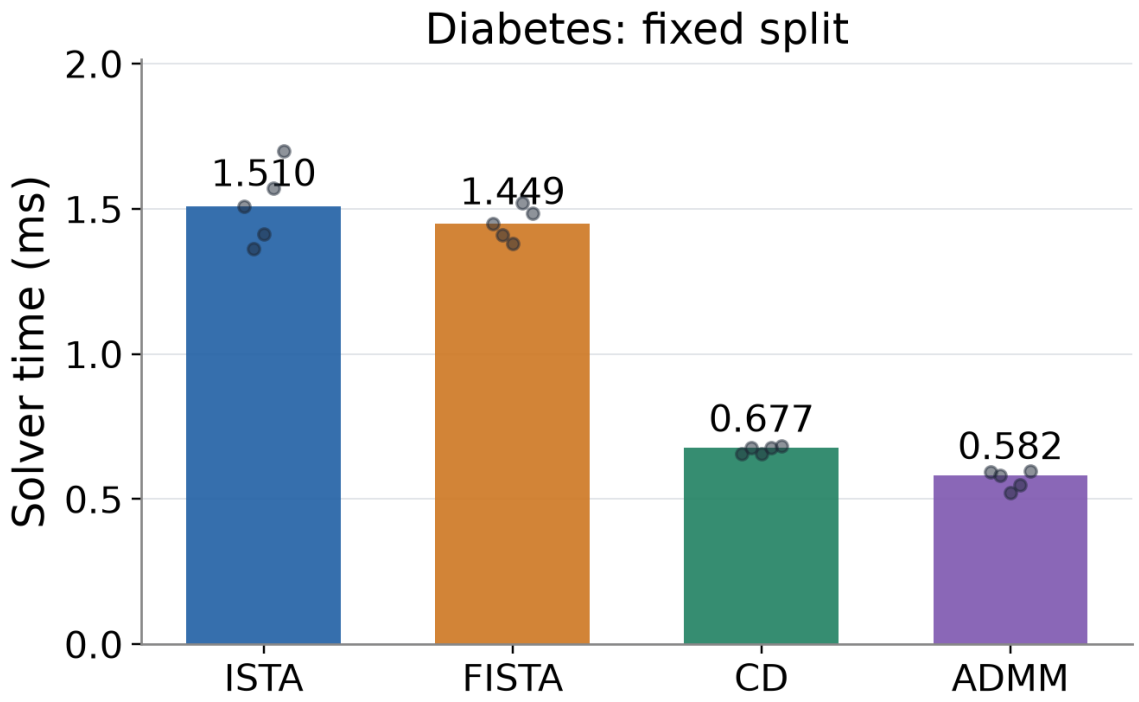
Tiny test-error differences at fixed λ provide no evidence of optimizer predictive superiority.

Report Section 5.2, Table 4; synthetic values are means ± sample SD across three instances.

Main Runtime Depends on the Dataset

ISTA has the lowest synthetic mean; ADMM is fastest on Diabetes here.



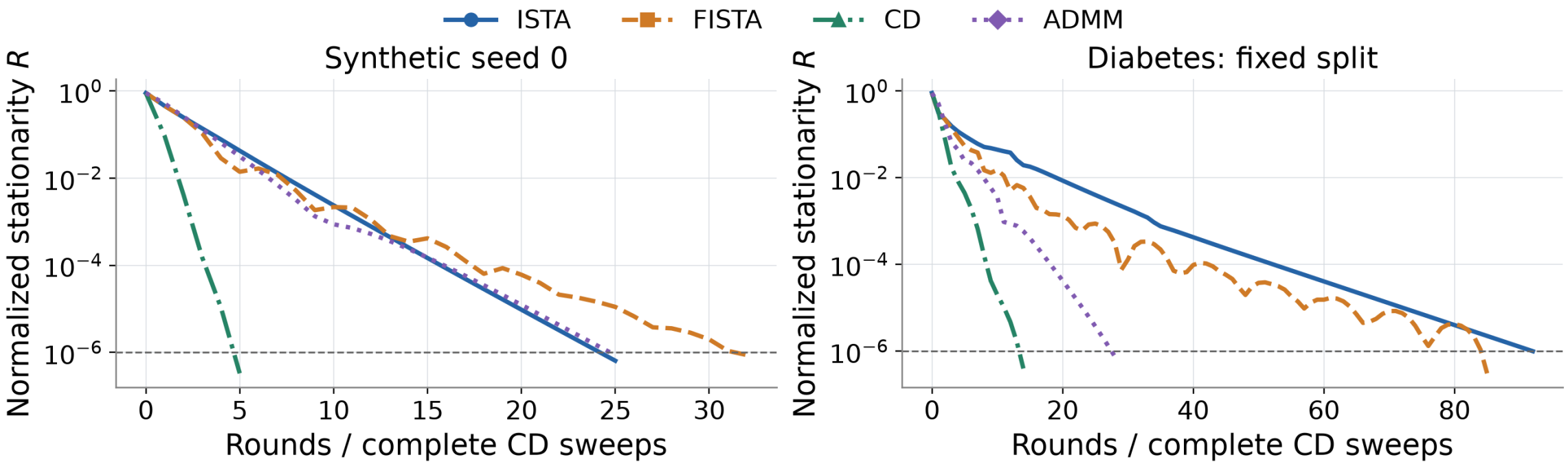


Time includes ADMM preparation. The conclusion concerns these implementations and this host.

Report Section 5.1, Table 3; 5 timings per instance. Synthetic: means ± sample SD of three instance medians.

Convergence at a Common Residual Target

The common residual shows when each solver reaches the accuracy target.

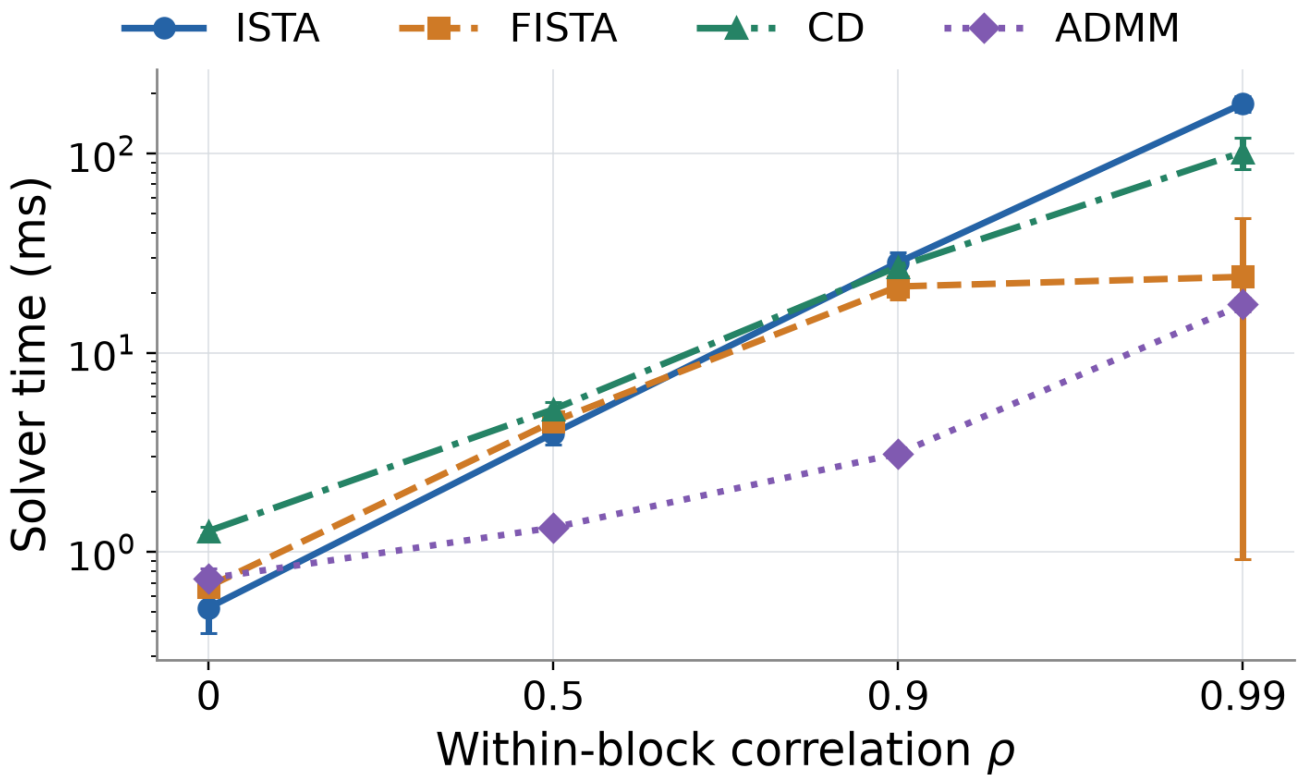


FISTA's accelerated objective bound does not guarantee earlier residual stopping on each problem.

Report Section 5.1; median-time histories. A CD round is a complete sweep; ADMM includes all split updates.

Strong Correlation Changes the Ranking

Strong correlation changes the ranking and exposes instance variation.



At ρ = 0.99: time (ms)

Seed	FISTA	ADMM
10	8.30	19.06
11	50.59	16.83
12	13.24	16.47

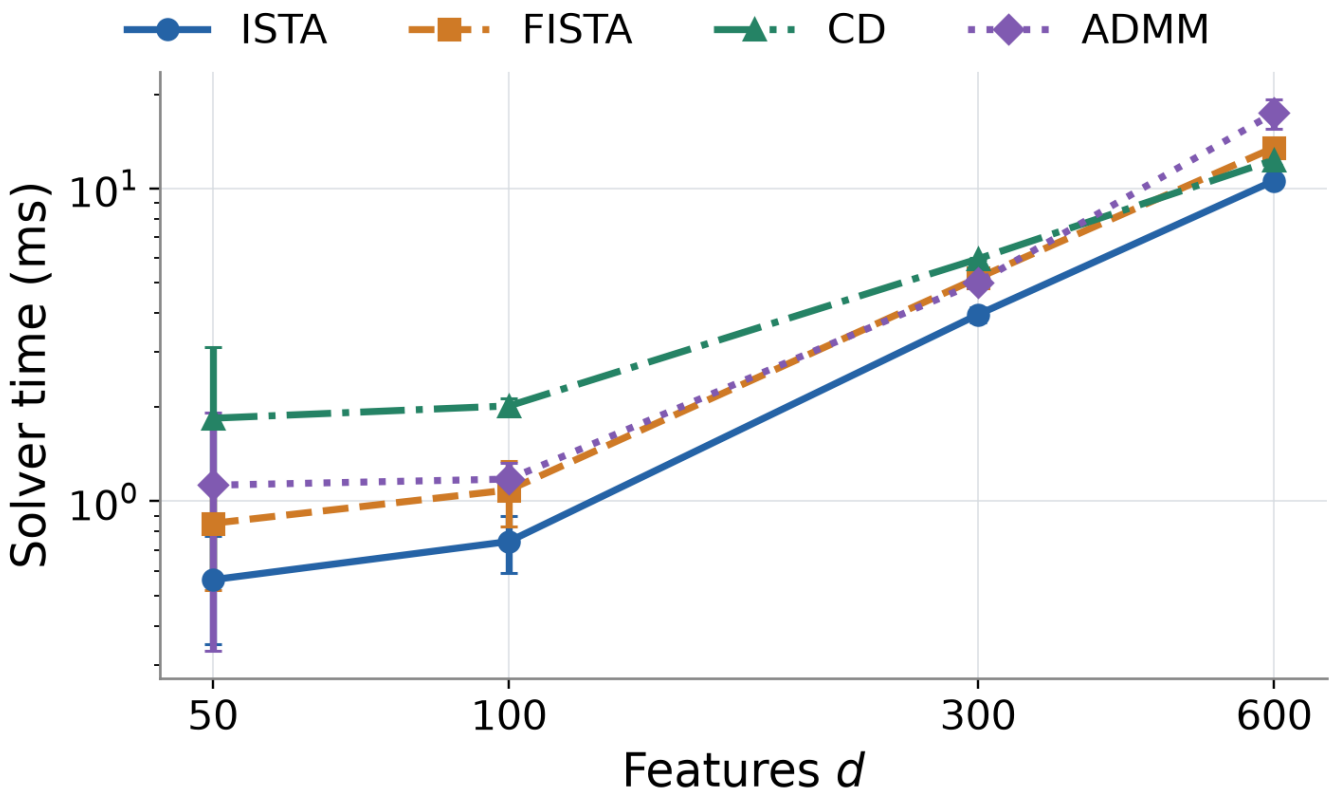
ADMM has the lower mean;
FISTA wins on 2 of 3 seeds.

At ρ = 0.99: ADMM 17.457 ± 1.401 ms; FISTA 24.041 ± 23.123 ms.

Report Section 5.4, Table 6; paired seeds 10–12. Error bars show sample SD across three instance medians.

More Features Increase Work and Setup Cost

ADMM's cold preparation becomes material as the feature count grows.



At d = 600

ADMM total: 17.413 ms

Preparation: 8.251 ms

47% of the total

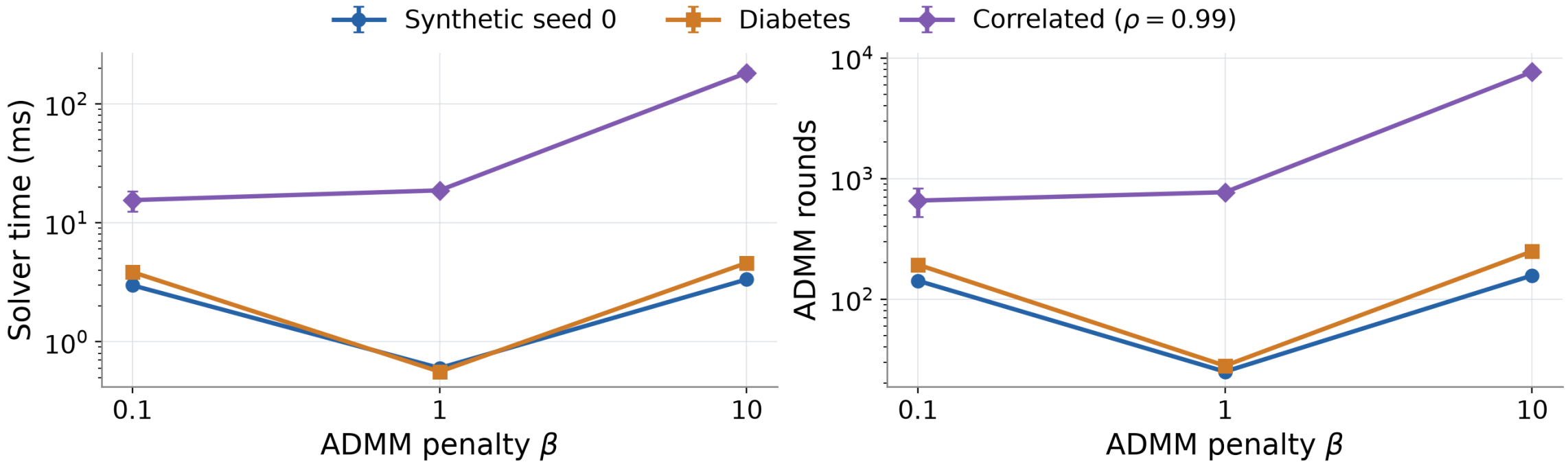
ISTA total: 10.582 ms

CD still uses 4 sweeps, but each sweep visits more coordinates; round count alone misses the cost.

Report Section 5.5 and Table 7; means over three representative median-time runs, 800 training samples.

ADMM's β Changes the Algorithm, Not the Model

ADMM's best algorithmic penalty depends on the input.

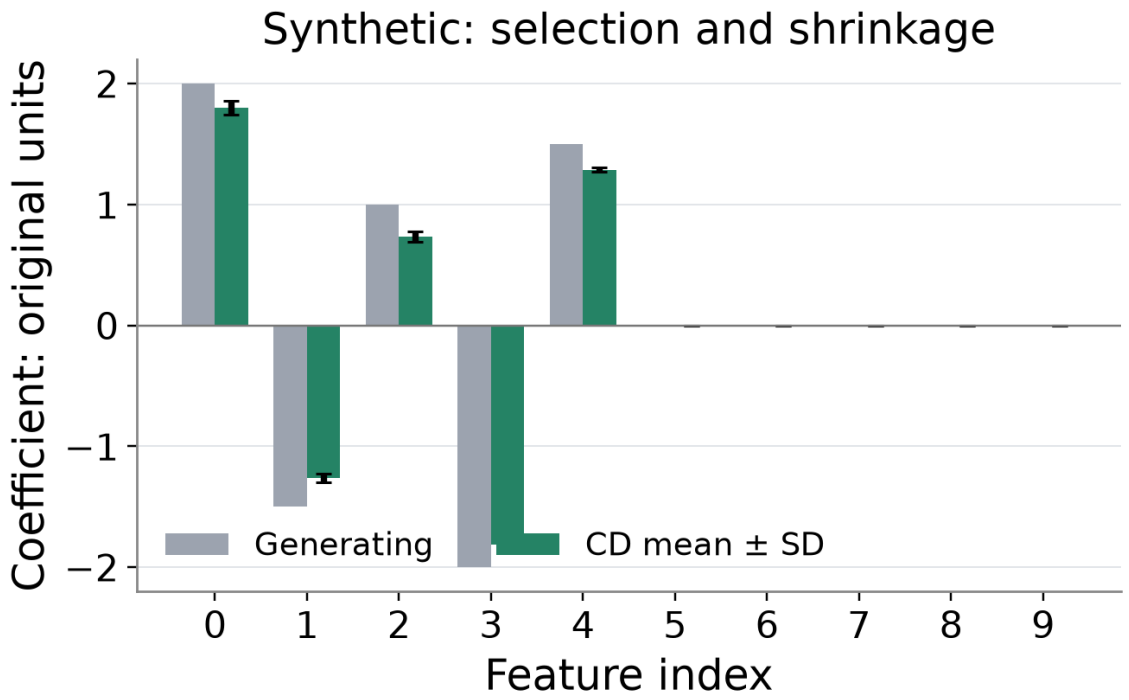


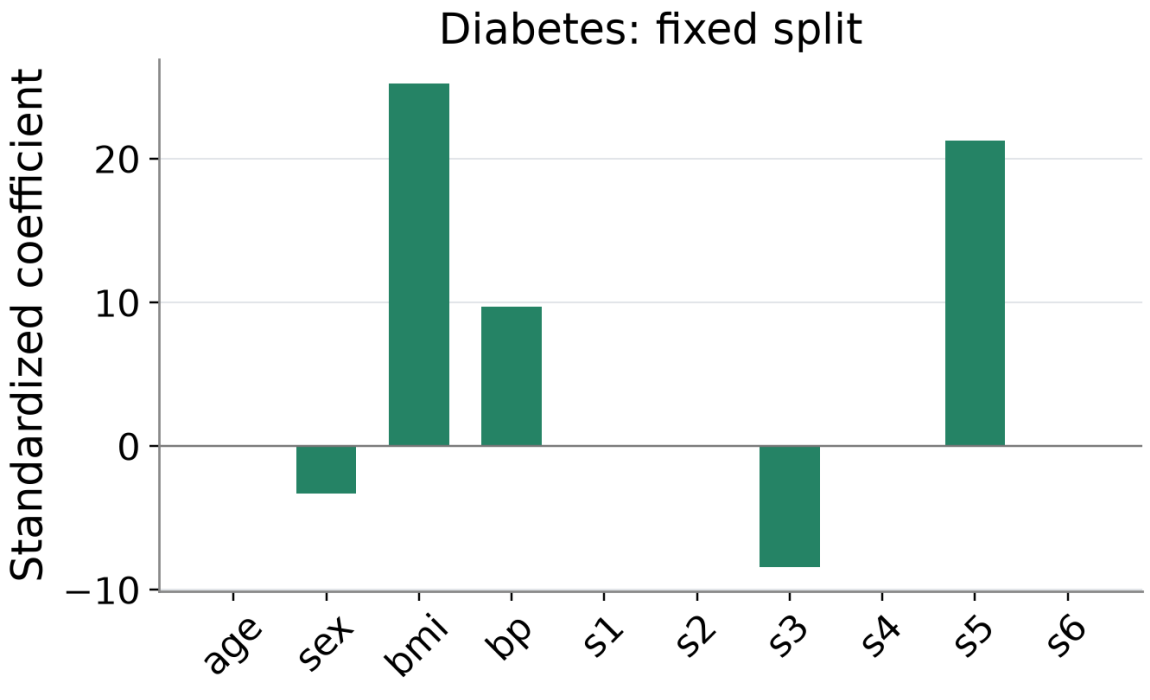
Model λ is fixed. The main comparison keeps predeclared β = 1 regardless of sensitivity outcomes.

Report Section 5.6, Table 8; β = 0.1, 1, 10. Three timings per setting; 45 runs in this separate study.

Correct Support Can Still Have Shrinkage Error

Exact support recovery still leaves shrinkage error in the retained coefficients.



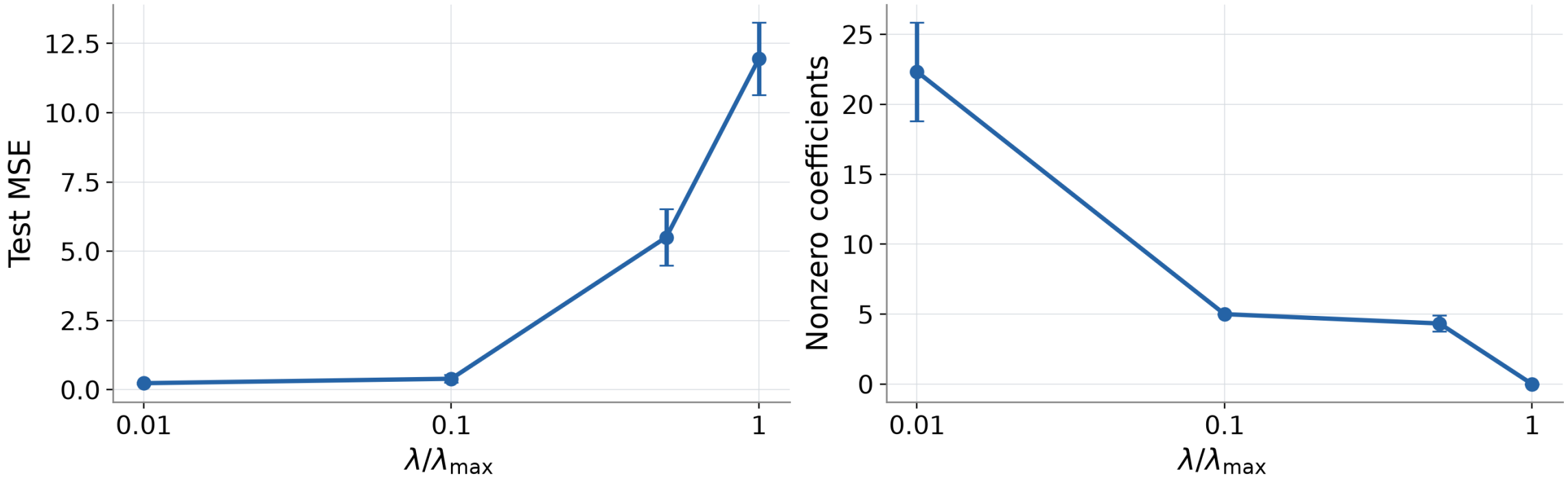


Synthetic support is known; Diabetes coefficients describe associations on one saved split.

Report Section 5.3; CD represents a converged solution. Synthetic coefficients are restored to original units.

The Model Penalty Trades Fit Against Sparsity

A smaller selected model need not have the smallest prediction error.



At λ/λmax = 0.1, the synthetic fits recover 5 variables but have higher average test MSE than at 0.01.

Report Section 5.3 and Table 5; CD, means across three synthetic instances. Test scores are not used for tuning.

Conclusions and Limits

The useful solver choice depends on design, implementation and accuracy.

Independent features

ISTA has the lowest mean time in our tests.

Positive correlation

ADMM has the lowest mean;
FISTA varies by instance.

Diabetes, one split

ADMM has the lowest recorded cold median.

Scope: small dense designs · 3 synthetic instances · 1 real-data split
Python/native-kernel and monitoring costs remain part of the comparison.

Model λ changes sparsity and shrinkage; algorithmic β changes ADMM's computation.

Report Section 6; conclusions are conditional on the recorded datasets, preprocessing, implementations and stopping rule.

Contributions and Discussion

Four implementations, shared writing and shared experimental evaluation.

Member	Student ID	Implementation	Presentation role
CAI Yangxiao	26023041R	ADMM	Part I
XUE Pengyu	26022951R	FISTA	Part II
ZHANG Kejia	26049131R	Cyclic CD	Opening + PPT design
ZHOU Xiyu	26025265R	ISTA	Part III

Report writing and revision: all four members

Experimental design and evaluation: all four members

Presentation design and opening overview: ZHANG Kejia. All four members join the Q&A.

Member responsibilities supplied and confirmed by Group 9; report Author contributions section.

Discussion

10-minute Q&A · All four members

CAI — formulation and ADMM

XUE — acceleration and comparison protocol

ZHANG — cyclic CD and reproducibility

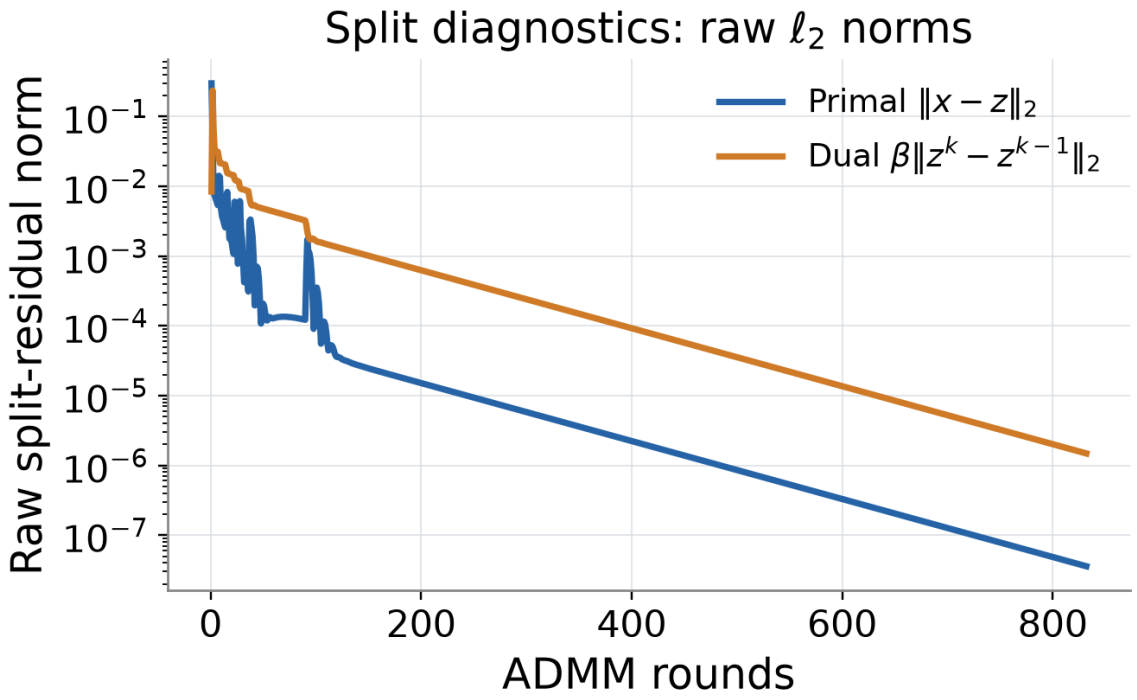
ZHOU — ISTA and empirical findings

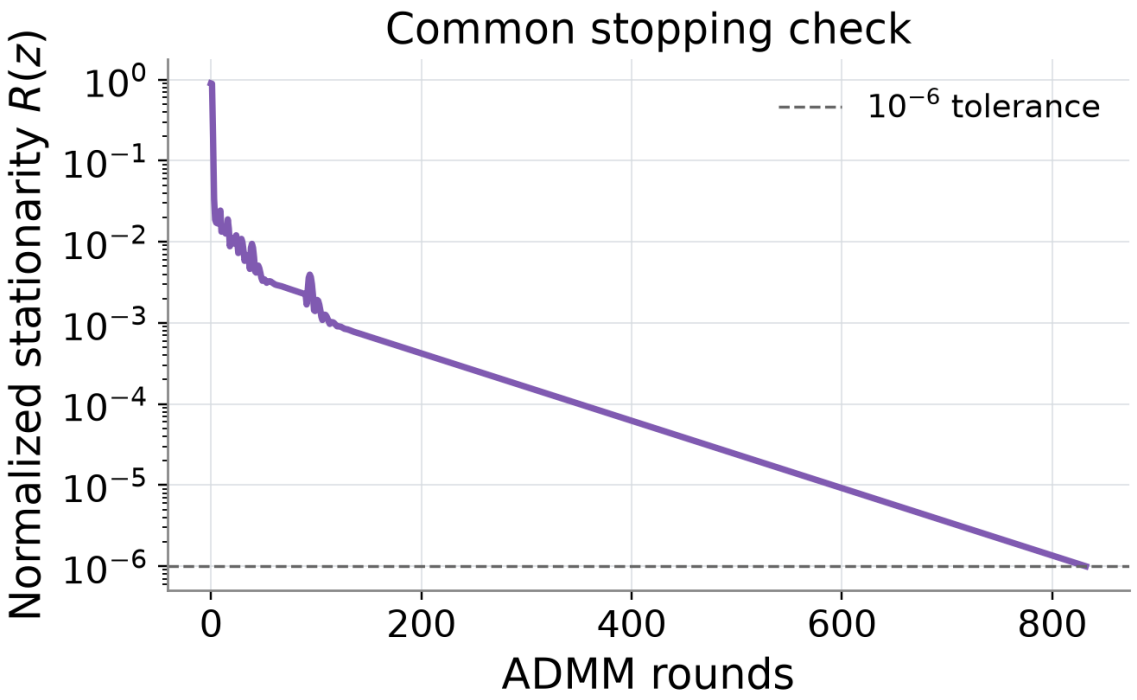


Code, data and reproducibility: github.com/Caiyangxiao/COMP6704-Group9-Project

Backup — ADMM Residual Diagnostics

ADMM's split diagnostics explain progress; common R(z) controls stopping.



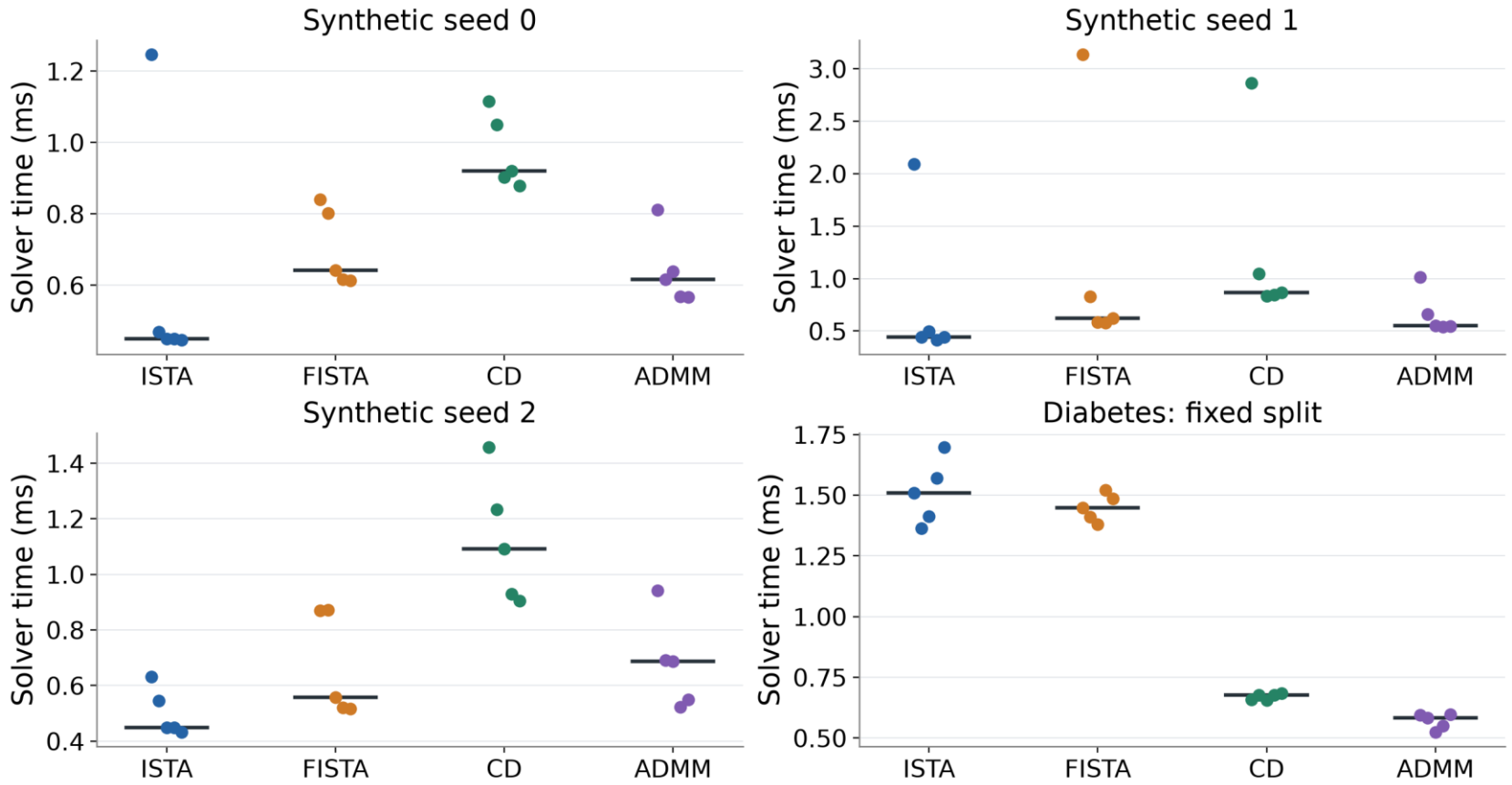


Primal: $\|x - z\|_2$. Dual: $\beta\|z^{k+1} - z^k\|_2$. These raw norms have a different scale from R(z).

Backup; saved ADMM diagnostics and common residual on correlation 0.99, seed 10. No extra stopping criterion is imposed.

Backup — Raw Timing Repetitions

The full repetitions retain outliers and overlapping timing ranges.



Five fixed-data repetitions support descriptive timing comparisons; no significance test is claimed.

Backup; main/timing_repeats.csv. Sample variation across data instances is a separate source of uncertainty.

Selected primary references and the reproducible project repository.

Tibshirani (1996)
Regression shrinkage and selection via the lasso.

Daubechies et al. (2004)
Iterative thresholding for linear inverse problems.

Beck & Teboulle (2009)
FISTA for linear inverse problems.

Friedman et al. (2010)
Coordinate descent regularization paths.

Wu & Lange (2008)
Coordinate descent for lasso-penalized regression.

Boyd et al. (2011)
ADMM for distributed optimization and learning.

Parikh & Boyd (2014)
Proximal algorithms.

Efron et al. (2004)
Least angle regression; Diabetes context.

Tseng (2001)
Block coordinate descent convergence.

Tibshirani (2013)
The lasso problem and uniqueness.

github.com/Caiyangxiao/COMP6704-Group9-Project

Backup; full bibliographic details and links appear in the report's References. All empirical figures use Group 9 saved results.